THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 41	LCM plus the common remainder
2	**(d)** 57	71 multiples up to 500 minus 14 multiples below 100
3	**(b)** 112	Count multiples from 100 to 999
4	**(c)** 111	Test palindromes 101, 111, 121, ...
5	**(d)** 0	Divisible by 10 and 9
6	**(a)** 5	Apply the routine and count
7	**(d)** 99	9 and 11 have no common factor
8	**(b)** 67	LCM plus the common remainder
9	**(c)** 59	75 multiples up to 900 minus 16 multiples below 200
10	**(c)** 69	Count multiples from 100 to 999

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	LCM(20, 25) $= 100$ (1); number $= 100 + 3 = 103$ (1).	1+1
12	72 multiples up to 800 (1); minus 13 below 150, giving 59 (1).	1+1
13	Middle number $= 45 \div 5 = 9$ (1); numbers 7, 8, 9, 10, 11 (1).	1+1
14	LCM(24, 36) $= 72$ (1); number $= 72 + 5 = 77$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$247247 = 247 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 35321, 22477, 19019) (1).	1+1+1
16	LCM $= 24$ (1); 9999 divided by it leaves remainder 15 (1); number 9984 (1).	1+1+1
17	LCM $= 24$ (1); multiples from 100 to 999: $41 - 4$ (1); 37 (1).	1+1+1
	OR: $358358 = 358 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 51194, 32578, 27566) (1).	1+1+1
18	LCM $= 60$ (1); 9999 divided by it leaves remainder 39 (1); number 9960 (1).	1+1+1
	OR: LCM $= 36$ (1); multiples from 100 to 999: $27 - 2$ (1); 25 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Splitting 45 into coprime factors and applying the rules (2); possibilities: a = 2, b = 0 (number 4230), a = 6, b = 5 (number 4635) (2); verification (1).	2+2+1
	OR: 832-238=594 (1); reverse of 594 is 495 (1); 594+495=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2
20	Splitting 90 into coprime factors and applying the rules (2); possibilities: a = 1, b = 0 (number 6120) (2); verification (1).	2+2+1
	OR: 641-146=495 (1); reverse of 495 is 594 (1); 495+594=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 212	Test palindromes 101, 111, 121, ...
2	**(b)** 4	Divisible by 10 and 9
3	**(a)** 5	Apply the routine and count
4	**(a)** 72	8 and 9 have no common factor
5	**(c)** 46	LCM plus the common remainder
6	**(d)** 63	66 multiples up to 1000 minus 3 multiples below 50
7	**(b)** 53	Count multiples from 100 to 999
8	**(b)** 222	Test palindromes 101, 111, 121, ...
9	**(b)** 5	Divisible by 10 and 9
10	**(c)** 7	Apply the routine and count

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	107 multiples up to 1500 (1); minus 17 below 250, giving 90 (1).	1+1
12	Middle number $= 80 \div 5 = 16$ (1); numbers 14, 15, 16, 17, 18 (1).	1+1
13	LCM(28, 42) $= 84$ (1); number $= 84 + 9 = 93$ (1).	1+1
14	100 multiples up to 600 (1); minus 12 below 75, giving 88 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$429429 = 429 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 61347, 39039, 33033) (1).	1+1+1
16	LCM $= 90$ (1); 9999 divided by it leaves remainder 9 (1); number 9990 (1).	1+1+1
17	LCM $= 70$ (1); multiples from 100 to 999: $14 - 1$ (1); 13 (1).	1+1+1
	OR: 516516 $= 516 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 73788, 46956, 39732) (1).	1+1+1
18	LCM $= 84$ (1); 9999 divided by it leaves remainder 3 (1); number 9996 (1).	1+1+1
	OR: LCM $= 60$ (1); multiples from 100 to 999: $16 - 1$ (1); 15 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Splitting 36 into coprime factors and applying the rules (2); possibilities: a = 0, b = 6 (number 5076), a = 4, b = 2 (number 5472), a = 9, b = 6 (number 5976) (2); verification (1).	2+2+1
	OR: 954-459=495 (1); reverse of 495 is 594 (1); 495+594=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2
20	Splitting 45 into coprime factors and applying the rules (2); possibilities: a = 0, b = 0 (number 8010), a = 4, b = 5 (number 8415), a = 9, b = 0 (number 8910) (2); verification (1).	2+2+1
	OR: 723-327=396 (1); reverse of 396 is 693 (1); 396+693=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 35	5 and 7 have no common factor
2	**(a)** 57	LCM plus the common remainder
3	**(c)** 69	80 multiples up to 2000 minus 11 multiples below 300
4	**(b)** 41	Count multiples from 100 to 999
5	**(c)** 161	Test palindromes 101, 111, 121, ...
6	**(c)** 0	Divisible by 10 and 9
7	**(c)** 7	Apply the routine and count
8	**(a)** 56	7 and 8 have no common factor
9	**(d)** 101	LCM plus the common remainder
10	**(b)** 99	111 multiples up to 999 minus 12 multiples below 111

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Middle number $= 115 \div 5 = 23$ (1); numbers 21, 22, 23, 24, 25 (1).	1+1
12	LCM(30, 45) $= 90$ (1); number $= 90 + 8 = 98$ (1).	1+1
13	40 multiples up to 5000 (1); minus 7 below 1000, giving 33 (1).	1+1
14	Middle number $= 150 \div 5 = 30$ (1); numbers 28, 29, 30, 31, 32 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$683683 = 683 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 97669, 62153, 52591) (1).	1+1+1
16	LCM $= 240$ (1); 9999 divided by it leaves remainder 159 (1); number 9840 (1).	1+1+1
17	LCM $= 84$ (1); multiples from 100 to 999: $11 - 1$ (1); 10 (1).	1+1+1
	OR: 791791 $= 791 \times 1001$ (1); $1001 = 7 \times 11 \times 13$ (1); so divisible by 7, 11 and 13 (quotients 113113, 71981, 60907) (1).	1+1+1
18	LCM $= 108$ (1); 9999 divided by it leaves remainder 63 (1); number 9936 (1).	1+1+1
	OR: LCM $= 144$ (1); multiples from 100 to 999: $6 - 0$ (1); 6 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Splitting 72 into coprime factors and applying the rules (2); possibilities: a = 0, b = 6 (number 3096) (2); verification (1).	2+2+1
	OR: 865-568=297 (1); reverse of 297 is 792 (1); 297+792=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2
20	Splitting 18 into coprime factors and applying the rules (2); possibilities: a = 0, b = 2 (number 2052), a = 2, b = 0 (number 2250), a = 3, b = 8 (number 2358), a = 5, b = 6 (number 2556), a = 7, b = 4 (number 2754), a = 9, b = 2 (number 2952) (2); verification (1).	2+2+1
	OR: 971-179=792 (1); reverse of 792 is 297 (1); 792+297=1089 (1); in general the difference is $99(a - c)$, whose digits are $(a - c - 1), 9, (10 - a + c)$ and the sum is always 1089 (2).	1+1+1+2

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